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Design Note 53 Nov. 78

77-68 32K - DYNARAM

This PCB accomodates two 16K bytes blocks of dynamic RAM, holds all the refresh control logic and provides 16 a clock generator for nominal data rates up to 4800 baud.

The following signals are ~~available on the 77-68 bus and form the interface with the CPU c~~ard. used to interface the memory board to the CPU board and are fully compatible with the 77/68 bus structure.

- R/$\overline{W}$
- Ø2
- E or VMA
- 5MHz
- Voltage supplies ± 12V and 5V

The edge connector designations are compatible with the 77-68 and ROM MONITOR PCB's.

An on board 8 pole switch allows positioning of each 16K byte block anywhere within the ~~16 bit~~ 64Kbyte. address space. An extra 5 control lines (active high) extend the address and/or allows memory to be disabled while accessing ROM or I/O anywhere within the 32K byte address space. For example this RAM could be used to 'fill-in' between MIKBUG, I/O and scratchpad locations.

Dynamic RAM's are employed with all the necessary mutliplexing and refresh logic provided - ~~no special purpose chips are required~~ by standard TTL L.S. chips. By disconnecting links 1 and 2 refresh may be exercised via the edge connectors to suit individual requirements.

When driven from the 77-68 bus the refresh cycle is 'hidden' that is it occurs during the first half of a Ø2 cycle with the advantage that there is no reduction in overall access time.

Explanatory Notes

Devices X4, 5, 13 and X12 multiplex the 14 bit address and the the 7 bit refresh address busses onto the 7 bit RAM address bus.

The multiplexer control signals (X11 pin 5 and X10 pin 9) select the CPU address bus and the refresh address bus respectively.

B $\overline{\emptyset 2}$, on X9 pin 13 conditions the first part of each $\emptyset 2$ cycle for refresh, if required. The remaining part of the cycle is reserved for normal read/write operations.

$\overline{\text{RAS}}$ the row address select occurs twice per $\emptyset 2$ cycle since it is required for both refresh and data read/write.

$\overline{\text{CAS}}$ - the construction of this waveform includes the chip select signal (X10 pins 16 and 9). It is not required during refresh and therefore occurs only during the latter half of a $\emptyset 2$ cycle.

Additional address decoding

Pads are provided on X18 pins 2-6 to be wired as required to any appropriate edge connector. These inputs may be used to provide an extension to the normal address space (64K) or to disable part of a 16K block that may already be used for ROM, RAM or I/O. By bringing any of these pins low the memory board will be deselected.

Write protection

One 16K byte block may be 'write protected' by bringing X18 pin 9 low. Pin 2 on diagram. Pads are provided to allow strapping to an edge connector or a switch.

Address Selection

switch posn '1' is furthest from the edge conn.

X10 SWITCH POSITION	ADDRESSES X23-X30	X10 SWITCH POSITION	ADDRESSES X31-X38
1	0000-3FFF	2	0000-3FFF
3	4000-7FFF	4	4000-7FFF
5	8000-BFFF	6	8000-BFFF
7	C000-FFFF	8	C000-FFFF

CONSTRUCTION

Connections and Links

Initially, do not connect to the additional decode inputs (X17 pins 2-6) or the write protect input (X18 pin 9).

Make all wire links, the holes are large enough to enable through links to be made by looping in and out of each hole.

Insert links 1 and 2.

Insert the transistor, resistors and capacitors and all RAM sockets (X23 - X38). Use a clean, small soldering iron bit and minimum amount of solder.

Examine the PCB carefully, with a magnifying glass if possible for solder bridges - most of your problems will be of this nature.

Check for power supply shorts.

Back Wiring

If a back plane is used isolate the clock outputs and the refresh control signals. Otherwise the connections are directly compatible with 77-68.

Insert the board and power up. Check for +12V, -5V and +5V at each RAM position - do not insert any I.C.'s yet.

Set the DIP switch (X10) to 1 and 4. This will position the RAM block (X23 - X30) in the address space 0000-3FFF (HEX) and the remaining block at 4000-7FFF.

Fill all I.C. positions except RAM, then power up.

Check that the counter chain is working by measuring approximately 2V on X8 pin 8 and X16 pin 12. On X22 pin 9 measure 0.25 - 0.5 volts.

Do not proceed further until all these conditions have been satisfied.

Inserting the 4116's

The substrate bias for these devices is supplied by the -5V rail and it is essential that no other pin on the device is ever subjected to a voltage more negative than this, that includes the period when the power supplies are switched on.

To some extent this has been safeguarded by slugging the remaining supplies relative to -5V. In fact it is essential that VBB is never more +ve than VSS If independant supplies are used, then the -12V supply must be switched on first and off last.

Bias failure

If the -5V supply fails the bleed resistor will keep the bias voltage close to ground but the safest approach is to use the -5V supply to switch the +5V and +12V supplies. To help with this approach the -5V supply has been taken out to an edge connector. Pin 73

Testing

Now observing the usual precautions when handling MOS insert just one 4116 in position X23 (the right way round).

Using MIKBUG or equivalent insert the following sequence of words at one or a number of addresses in the range 0000 to 3FFF.

INPUT 0000,0000

RESULT 0XXX,XXXX   'X' = don't care state.

INPUT 1111,1111

RESULT 1XXX,XXXX

Move the device from X23 to X24 then

INPUT 0000,0000

RESULT X0XX,XXXX

INSERT 1111,1111

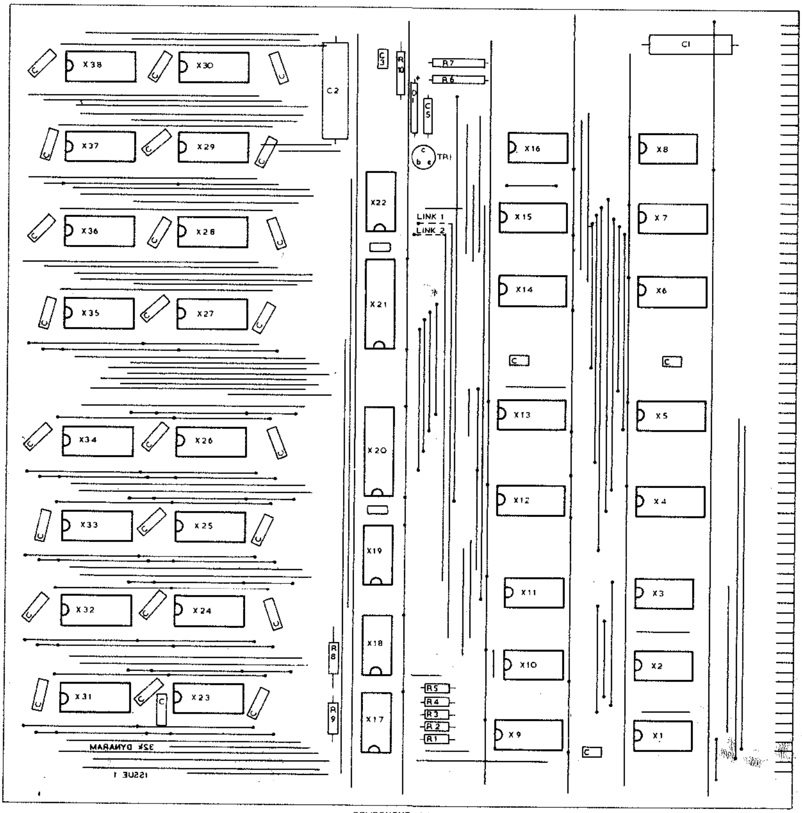
RESULT X1XX,XXXX

Then move the device from X24 to X25 and so on until the complete byte has been checked. Now change the address selection switch X10 so that 4 is on and continue as above with the second block of memory. (4000-7FFF).

Once this has all checked out insert all the RAM chips and run the memory test programme attached.

Components DYNARAM Dec. 78

X12, X13, X4, X5	74LS153
X23-X38	~~MB~~4116 ~~(NE or H)~~
X20, X21	DM81LS97 or DM81LS95
X9	74LS139
X1, X8, X16	74LS04
X2, X22, X11	74LS74
X19	74LS32
X14, X6, X7, X15	74LS163
X17	74LS30
X18	74LS00
D1	5.6V Zener diode
TR1	BCY70
R1, R2, R3, R4, R5, R6, R8, 9, 10	2.7K ~~(TR4)~~
R7	30.
C1	ELECTROLYTIC 100uf 10V
C2	100uf 20V
X10 switch	8 Pole single throw DIP
All other capacitors (31 total)	0.1uf ceramic disc



COMPONENT SIDE

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32 kbyte 77-68 RAM BOARD